

Java Notes - Memory Management (Stack & Heap)

1. Memory Management

Java manages memory using two memory areas: **Stack Memory** and **Heap Memory**.

2. Stack Memory

- Follows **LIFO (Last In, First Out)**.
- Stores local variables, method parameters, reference variables, and stack frames.
- Memory is released automatically when a method finishes.

3. Heap Memory

- Stores objects created using the **new** keyword.
- Objects contain instance variables.
- Unused objects are removed by the Garbage Collector.

4. Example

```
class Student {  
    int instVariable;  
    public static void main(String[] args){  
        int localVariable = 0;  
        Student st = new Student();  
    }  
}
```

Variable Allocation

- **Instance Variable** → Stored in Heap.
- **Local Variable** → Stored in Stack.
- **Reference Variable (st)** → Stored in Stack and points to an object in Heap.

Stack vs Heap

Stack: Local variables, reference variables, method calls.

Heap: Objects and instance variables.

Quick Revision

Stack = Local Variables + References + Method Calls

Heap = Objects + Instance Variables

new = Creates Object in Heap

Garbage Collector = Removes unused objects